SQL

157 questions · compiled from 3 files

SQL BASICS

1. Which SQL statement is used to retrieve data from a database table?

1. GET
2. OPEN
3. SELECT
4. EXTRACT

Answer 3 · SELECT

SQL BASICS

2. Which clause specifies the table to read data from?

1. WHERE
2. FROM
3. ORDER BY
4. GROUP BY

Answer 2 · FROM

SQL BASICS

3. What does the asterisk (*) mean in SELECT * FROM employees?

1. Multiply all columns
2. Select all rows twice
3. Select all columns
4. Select only numeric columns

Answer 3 · Select all columns

LIMIT

4. Which clause is used to restrict the number of rows returned?

1. LIMIT
2. STOP
3. TOPONLY
4. COUNT

Answer 1 · LIMIT

ORDER BY

5. Which clause is used to sort query results?

1. GROUP BY
2. ORDER BY
3. SORT
4. ARRANGE BY

Answer 2 · ORDER BY

ORDER BY

6. **What is the default sort order in ORDER BY?**

   1. DESC
   2. RANDOM
   3. ASC
   4. NONE

   Answer 3 · ASC

WHERE

7. **Which clause is used to filter rows based on a condition?**

   1. WHERE
   2. FILTER
   3. HAVING
   4. CHECK

   Answer 1 · WHERE

WHERE

8. **Which operator checks if a value is equal to another value?**

   1. ==
   2. =
   3. EQUALS
   4. IS SAME

   Answer 2 · =

ARITHMETIC OPERATORS

9. **Which expression increases salary by 1000 in the result?**

   1. salary ++ 1000
   2. salary + 1000
   3. ADD(salary,1000)
   4. salary PLUS 1000

   Answer 2 · salary + 1000

LIKE

10. **Which operator is used for pattern matching in SQL?**

    1. MATCH
    2. SEARCH
    3. LIKE
    4. FIND

    Answer 3 · LIKE

LIKE

11. **Which wildcard represents any number of characters in a LIKE pattern?**

    1. _
    2. *
    3. %
    4. #

    Answer 3 · %

LIKE

12. Which wildcard represents exactly one character in a LIKE pattern?

1. ?
2. _
3. %
4. *

Answer 2 · _

IN

13. Which operator is used to match a value against a list of values?

1. BETWEEN
2. IN
3. AMONG
4. LIST

Answer 2 · IN

BETWEEN

14. Which operator checks whether a value falls within a range?

1. RANGE
2. INSIDE
3. BETWEEN
4. WITHIN

Answer 3 · BETWEEN

AND OR

15. Which operator returns rows only when both conditions are true?

1. OR
2. AND
3. BOTH
4. ALL

Answer 2 · AND

AND OR

16. Which operator returns rows when at least one condition is true?

1. OR
2. AND
3. ANY
4. EITHER

Answer 1 · OR

NOT

17. Which operator reverses a condition?

1. NO
2. REVERSE
3. NOT
4. !

Answer 3 · NOT

JOIN BASICS

18. What is the purpose of a JOIN in SQL?

1. To delete rows
2. To combine rows from related tables
3. To sort data
4. To rename columns

Answer 2 · To combine rows from related tables

INNER JOIN

19. Which JOIN returns only matching rows from both tables?

1. LEFT JOIN
2. FULL JOIN
3. INNER JOIN
4. CROSS JOIN

Answer 3 · INNER JOIN

LEFT JOIN

20. Which JOIN returns all rows from the left table and matching rows from the right table?

1. RIGHT JOIN
2. LEFT JOIN
3. INNER JOIN
4. SELF JOIN

Answer 2 · LEFT JOIN

COUNT

21. Which aggregate function counts rows?

1. SUM
2. TOTAL
3. COUNT
4. NUMBER

Answer 3 · COUNT

SUM

22. Which aggregate function adds numeric values?

1. ADD
2. SUM
3. TOTAL
4. COUNT

Answer 2 · SUM

AVG

23. Which aggregate function returns the average of numeric values?

1. MEAN
2. AVG
3. AVERAGE
4. MID

Answer 2 · AVG

MIN MAX

24. Which function returns the smallest value in a column?

1. LOWEST
2. MIN
3. SMALLEST
4. FIRST

Answer 2 · MIN

MIN MAX

25. Which function returns the largest value in a column?

1. HIGH
2. TOP
3. MAX
4. LARGEST

Answer 3 · MAX

GROUP BY

26. Which clause groups rows that have the same values in specified columns?

1. ORDER BY
2. GROUP BY
3. COLLECT BY
4. HAVING

Answer 2 · GROUP BY

HAVING

27. Which clause filters grouped results after aggregation?

1. WHERE
2. HAVING
3. FILTER
4. ORDER BY

Answer 2 · HAVING

DISTINCT

28. What does DISTINCT do in a SELECT statement?

1. Sorts rows
2. Removes duplicate rows from the result
3. Counts rows
4. Groups rows

Answer 2 · Removes duplicate rows from the result

NULL BASICS

29. What does NULL represent in SQL?

1. Zero
2. An empty string
3. Missing or unknown value
4. False

Answer 3 · Missing or unknown value

NULL BASICS

30. Which expression correctly checks for NULL values?

1. = NULL
2. IS NULL
3. == NULL
4. NULL ==

Answer 2 · IS NULL

CASE

31. What is CASE used for in SQL?

1. Creating tables
2. Conditional logic in queries
3. Joining tables
4. Sorting rows

Answer 2 · Conditional logic in queries

SUBQUERIES

32. What is a subquery?

1. A backup table
2. A query inside another query
3. A temporary database
4. A duplicate row

Answer 2 · A query inside another query

CTES

33. What does CTE stand for?

1. Common Table Expression
2. Conditional Table Entry
3. Computed Table Element
4. Common Typed Entity

Answer 1 · Common Table Expression

TEMP TABLES

34. What is a temporary table mainly used for?

1. Permanent storage
2. Intermediate results during a session
3. Replacing indexes
4. User authentication

Answer 2 · Intermediate results during a session

WINDOW FUNCTIONS

35. What is a window function designed to do?

1. Delete duplicate rows
2. Perform calculations across related rows without collapsing them
3. Rename columns
4. Create indexes

Answer 2 · Perform calculations across related rows without collapsing them

WINDOW FUNCTIONS

36. Which clause is commonly used inside a window function to define row grouping?

1. HAVING
2. PARTITION BY
3. LIMIT
4. DISTINCT

Answer 2 · PARTITION BY

DATE FUNCTIONS

37. Which type of SQL function is used to extract year or month from a date?

1. String function
2. Date function
3. Join function
4. Aggregate function

Answer 2 · Date function

DATA CLEANING

38. Which SQL function is often used to replace NULL values with a default value?

1. COALESCE
2. REPLACE ALL
3. FILLNULL
4. DEFAULT

Answer 1 · COALESCE

PERFORMANCE

39. Why is SELECT * often discouraged in production queries?

1. It always causes errors
2. It may retrieve unnecessary columns
3. It removes duplicates
4. It prevents filtering

Answer 2 · It may retrieve unnecessary columns

ORDER BY

40. Which query correctly sorts salary from highest to lowest?

1. ORDER BY salary
2. ORDER BY salary DESC
3. SORT salary DOWN
4. GROUP BY salary DESC

Answer 2 · ORDER BY salary DESC

WHERE

41. Which condition returns rows where age is greater than 30?

1. age => 30
2. age > 30
3. age AFTER 30
4. age GREATER 30

Answer 2 · age > 30

LIKE

42. Which pattern matches names that start with 'A'?

1. %A
2. A%
3. _A%
4. %A%

Answer 2 · A%

IN

43. Which condition correctly finds departments that are either HR or Sales?

1. department = HR OR Sales
2. department IN ('HR','Sales')
3. department BETWEEN 'HR' AND 'Sales'
4. department LIKE 'HR,Sales'

Answer 2 · department IN ('HR','Sales')

BETWEEN

44. Which condition correctly finds salaries from 40000 to 60000 inclusive?

1. salary > 40000 AND salary < 60000
2. salary BETWEEN 40000 AND 60000
3. salary IN 40000 TO 60000
4. salary RANGE 40000,60000

Answer 2 · salary BETWEEN 40000 AND 60000

AND OR

45. Which condition finds rows where city is Delhi and age is above 25?

1. city = 'Delhi' OR age > 25
2. city = 'Delhi' AND age > 25
3. city = Delhi, age > 25
4. city AND age > 25

Answer 2 · city = 'Delhi' AND age > 25

NOT

46. Which condition excludes rows where status is 'Closed'?

1. status != 'Closed'
2. NOT status = 'Closed'
3. status <> 'Closed'
4. All of the above

Answer 4 · All of the above

INNER JOIN

47. Which clause defines how two tables are matched in a JOIN?

1. WHERE
2. MATCH BY
3. ON
4. USING ONLY

Answer 3 · ON

LEFT JOIN

48. If a LEFT JOIN finds no match in the right table, what is returned for right-table columns?

1. 0
2. Blank spaces
3. NULL
4. The left-table value

Answer 3 · NULL

GROUP BY

49. Why is GROUP BY used with aggregate functions?

1. To rename columns
2. To calculate values per group
3. To filter rows before selection
4. To sort grouped rows automatically

Answer 2 · To calculate values per group

HAVING

50. Why can't WHERE usually replace HAVING for aggregate filters?

1. WHERE runs after ORDER BY
2. WHERE filters rows before grouping, HAVING filters groups after aggregation
3. WHERE only works on text
4. HAVING is faster

Answer 2 · WHERE filters rows before grouping, HAVING filters groups after aggregation

DISTINCT

51. What does SELECT DISTINCT city FROM customers return?

1. All city values including duplicates
2. Unique city values only
3. Only the first city
4. Cities sorted descending

Answer 2 · Unique city values only

CASE

52. What will CASE typically return?

1. A table
2. A calculated value based on conditions
3. A new database
4. A join result only

Answer 2 · A calculated value based on conditions

SUBQUERIES

53. Where can a subquery commonly appear?

1. Only in SELECT
2. Only in FROM
3. Only in WHERE
4. In SELECT, FROM, or WHERE

Answer 4 · In SELECT, FROM, or WHERE

CTES

54. What is one advantage of a CTE?

1. It permanently stores results
2. It improves readability of complex queries
3. It replaces all joins
4. It removes NULL automatically

Answer 2 · It improves readability of complex queries

WINDOW FUNCTIONS

55. How is a window function different from GROUP BY?

1. It removes duplicate rows
2. It keeps individual rows while adding calculated values across related rows
3. It only works with dates
4. It cannot use aggregates

Answer 2 · It keeps individual rows while adding calculated values across related rows

PERFORMANCE

56. Which practice can improve query readability and maintainability?

1. Using meaningful aliases and formatting
2. Always using SELECT *
3. Avoiding indentation
4. Removing all comments

Answer 1 · Using meaningful aliases and formatting

SQL BASICS

57. Which query returns only the name and salary columns from an employees table?

1. SELECT employees FROM name salary
2. SELECT name salary FROM employees
3. SELECT name, salary FROM employees
4. GET name, salary IN employees

Answer 3 · SELECT name, salary FROM employees

WHERE

58. Which condition correctly filters rows where salary is less than or equal to 50000?

1. salary <= 50000
2. salary =< 50000
3. salary <=> 50000
4. salary LESS 50000

Answer 1 · salary <= 50000

ORDER BY

59. Which query sorts products first by category ascending and then by price descending?

1. ORDER BY category, price
2. ORDER BY category ASC, price DESC
3. SORT BY category ASC THEN price DESC
4. GROUP BY category ASC, price DESC

Answer 2 · ORDER BY category ASC, price DESC

LIMIT

60. Which query returns the first 10 rows from a table named orders?

1. SELECT 10 FROM orders
2. SELECT * FROM orders LIMIT 10
3. SELECT LIMIT 10 * FROM orders
4. GET TOP 10 FROM orders

*Answer 2 · SELECT * FROM orders LIMIT 10*

LIKE

61. Which pattern matches any five-character string that starts with A?

1. A%
2. A____
3. A_____
4. _A___

Answer 2 · A____

LIKE

62. Which pattern matches values ending with 'son'?

1. son%
2. %son
3. %son%
4. _son

Answer 2 · %son

IN

63. Which condition is the clearest way to filter rows where department is Sales, HR, or IT?

1. department = 'Sales' OR 'HR' OR 'IT'
2. department IN ('Sales','HR','IT')
3. department BETWEEN 'Sales' AND 'IT'
4. department LIKE 'Sales,HR,IT'

Answer 2 · department IN ('Sales','HR','IT')

BETWEEN

64. Which condition finds orders placed between January 1 and January 31 inclusive?

1. order_date > '2024-01-01' AND order_date < '2024-01-31'
2. order_date BETWEEN '2024-01-01' AND '2024-01-31'
3. order_date IN January
4. order_date LIKE '2024-01%'

Answer 2 · order_date BETWEEN '2024-01-01' AND '2024-01-31'

AND OR

65. What is the main reason parentheses are useful in complex WHERE clauses?

1. They rename columns
2. They control logical evaluation order
3. They speed up all queries automatically
4. They replace HAVING

Answer 2 · They control logical evaluation order

NOT

66. Which condition excludes rows where city is either Delhi or Mumbai?

1. city NOT IN ('Delhi','Mumbai')
2. city != 'Delhi','Mumbai'
3. NOT city = 'Delhi' OR 'Mumbai'
4. city EXCEPT ('Delhi','Mumbai')

Answer 1 · city NOT IN ('Delhi','Mumbai')

JOIN BASICS

67. If two tables share customer_id, what is the most likely purpose of joining on that column?

1. To sort by customer_id
2. To relate customer records to associated records in the other table
3. To remove duplicates automatically
4. To calculate averages only

Answer 2 · To relate customer records to associated records in the other table

INNER JOIN

68. Which query structure correctly represents an INNER JOIN?

1. SELECT * FROM a INNER JOIN b ON a.id = b.id
2. SELECT * FROM a JOIN WHERE a.id = b.id
3. SELECT * FROM a MATCH b USING id
4. SELECT * FROM a WITH b ON id

*Answer 1 · SELECT * FROM a INNER JOIN b ON a.id = b.id*

LEFT JOIN

69. What does a LEFT JOIN return?

1. Only matching rows
2. All rows from the right table and matching rows from the left
3. All rows from the left table and matching rows from the right
4. Only non-null rows

Answer 3 · All rows from the left table and matching rows from the right

LEFT JOIN

70. If no matching row exists in the right table during a LEFT JOIN, what appears for right-table columns?

1. 0
2. An error
3. NULL values
4. The left-table values

Answer 3 · NULL values

FULL OUTER JOIN

71. What is the result of a FULL OUTER JOIN?

1. Only matching rows
2. All rows from both tables with matches where available
3. Only rows from the left table
4. Only rows from the right table

Answer 2 · All rows from both tables with matches where available

SELF JOIN

72. When is a self join useful?

1. When comparing rows within the same table
2. When deleting a table
3. When creating indexes
4. When replacing GROUP BY

Answer 1 · When comparing rows within the same table

UNION

73. What does UNION do?

1. Joins tables side by side
2. Combines result sets vertically and removes duplicates
3. Counts rows from two tables
4. Creates a temp table

Answer 2 · Combines result sets vertically and removes duplicates

UNION ALL

74. How does UNION ALL differ from UNION?

1. It sorts automatically
2. It keeps duplicates
3. It only works on numbers
4. It requires a JOIN condition

Answer 2 · It keeps duplicates

COUNT

75. What does COUNT(DISTINCT customer_id) return?

1. Total rows in the table
2. Number of unique non-null customer_id values
3. Number of duplicated customer_id values only
4. Number of columns in the table

Answer 2 · Number of unique non-null customer_id values

SUM

76. What happens if SUM is applied to a numeric column containing NULLs?

1. The query fails
2. NULLs are ignored in the sum
3. NULLs are treated as text
4. NULLs are counted as 1

Answer 2 · NULLs are ignored in the sum

AVG

77. Why might AVG(score) differ from SUM(score)/COUNT(*)?

1. AVG ignores NULLs while COUNT(*) includes all rows
2. AVG sorts scores first
3. AVG rounds automatically to integers
4. There is never any difference

Answer 1 · AVG ignores NULLs while COUNT(*) includes all rows

GROUP BY

78. Which query is valid when using GROUP BY department?

1. SELECT department, AVG(salary) FROM employees GROUP BY department
2. SELECT department, salary FROM employees GROUP BY department
3. SELECT AVG(salary) FROM employees WHERE GROUP BY department
4. SELECT department FROM employees HAVING AVG(salary)

Answer 1 · SELECT department, AVG(salary) FROM employees GROUP BY department

HAVING

79. Which query correctly filters departments with an average salary above 60000?

1. SELECT department, AVG(salary) FROM employees WHERE AVG(salary) > 60000 GROUP BY department
2. SELECT department, AVG(salary) FROM employees GROUP BY department HAVING AVG(salary) > 60000
3. SELECT department FROM employees HAVING salary > 60000
4. SELECT AVG(salary) HAVING department > 60000

Answer 2 · SELECT department, AVG(salary) FROM employees GROUP BY department HAVING AVG(salary) > 60000

DISTINCT

80. Which statement about DISTINCT is true?

1. It always sorts results
2. It removes duplicate rows from the selected output columns
3. It only works with numbers
4. It is the same as GROUP BY in every case

Answer 2 · It removes duplicate rows from the selected output columns

CASE

81. Which CASE expression correctly labels salaries above 70000 as High and others as Standard?

1. CASE salary > 70000 THEN 'High' ELSE 'Standard' END
2. CASE WHEN salary > 70000 THEN 'High' ELSE 'Standard' END
3. WHEN salary > 70000 CASE 'High' ELSE 'Standard'
4. IF salary > 70000 THEN 'High'

Answer 2 · CASE WHEN salary > 70000 THEN 'High' ELSE 'Standard' END

CASE

82. Why is CASE useful in reporting queries?

1. It allows conditional categorization without changing stored data
2. It permanently updates the table
3. It replaces joins
4. It removes NULLs automatically

Answer 1 · It allows conditional categorization without changing stored data

DATE

83. Which task is a common use of date functions?

1. Grouping sales by month
2. Joining unrelated tables
3. Replacing primary keys
4. Removing all duplicates

Answer 1 · Grouping sales by month

SUBQUERIES

84. Which example best fits a subquery in the FROM clause?

1. Using a derived table for further aggregation
2. Filtering one row by ID
3. Sorting a single column
4. Renaming a table permanently

Answer 1 · Using a derived table for further aggregation

SUBQUERIES

85. Why can correlated subqueries be slower than simpler alternatives?

1. They may execute logic repeatedly for many outer rows
2. They cannot use WHERE
3. They always create temp tables on disk
4. They remove indexes

Answer 1 · They may execute logic repeatedly for many outer rows

CTES

86. What is one reason to choose a CTE over a deeply nested subquery?

1. It can make complex logic easier to read and debug
2. It always runs faster
3. It stores data permanently
4. It avoids all aggregations

Answer 1 · It can make complex logic easier to read and debug

CTES

87. What is a recursive CTE commonly used for?

1. Simple sorting only
2. Hierarchical or iterative data traversal
3. Replacing DISTINCT
4. Creating indexes

Answer 2 · Hierarchical or iterative data traversal

TEMP TABLES

88. Which scenario best fits a temporary table?

1. You need an intermediate dataset reused across several later queries
2. You need to rename one column
3. You want to sort one result once
4. You need to count one column only

Answer 1 · You need an intermediate dataset reused across several later queries

WINDOW FUNCTIONS

89. What does ROW_NUMBER() typically do?

1. Counts total rows in a table
2. Assigns a sequential number to rows within an ordered window
3. Returns only unique rows
4. Calculates an average

Answer 2 · Assigns a sequential number to rows within an ordered window

WINDOW FUNCTIONS

90. Which function would you use to rank rows while allowing ties to share the same rank with gaps after ties?

1. ROW_NUMBER()
2. RANK()
3. COUNT()
4. SUM()

Answer 2 · RANK()

WINDOW FUNCTIONS

91. Which function would you use to rank rows while allowing ties but without gaps in subsequent ranks?

1. DENSE_RANK()
2. RANK()
3. ROW_NUMBER()
4. AVG()

Answer 1 · DENSE_RANK()

WINDOW FUNCTIONS

92. Which expression is most associated with a running total?

1. SUM(amount) OVER (ORDER BY order_date)
2. COUNT(*) GROUP BY order_date
3. AVG(amount) WHERE order_date
4. SUM(amount) DISTINCT

Answer 1 · SUM(amount) OVER (ORDER BY order_date)

PARTITION BY

93. What is the effect of PARTITION BY department in a window function?

1. It filters departments
2. It restarts the window calculation for each department
3. It sorts departments alphabetically
4. It removes duplicates

Answer 2 · It restarts the window calculation for each department

WINDOW FUNCTIONS

94. Why can't a simple GROUP BY replace every window function use case?

1. GROUP BY removes row-level detail while window functions preserve it
2. GROUP BY is invalid with numbers
3. Window functions cannot aggregate
4. GROUP BY only works on text

Answer 1 · GROUP BY removes row-level detail while window functions preserve it

DATA CLEANING

95. Which SQL technique is commonly used to replace NULL values during cleaning?

1. COALESCE or similar null-handling logic
2. ORDER BY DESC
3. INNER JOIN only
4. LIMIT 1

Answer 1 · COALESCE or similar null-handling logic

DATA CLEANING

96. Which issue can DISTINCT sometimes help identify or reduce?

1. Duplicate rows
2. Missing indexes
3. Foreign key creation
4. Date formatting only

Answer 1 · Duplicate rows

DATA CLEANING

97. Why should analysts be careful when removing duplicates?

1. Some repeated rows may represent legitimate repeated events
2. Duplicates are always errors
3. Removing duplicates always improves accuracy
4. Duplicates only occur in text columns

Answer 1 · Some repeated rows may represent legitimate repeated events

FULL OUTER JOIN

98. When is a FULL OUTER JOIN especially useful?

1. When comparing two datasets to find matches and mismatches on both sides
2. When sorting one table
3. When calculating one average
4. When filtering one column

Answer 1 · When comparing two datasets to find matches and mismatches on both sides

JOINS WITH COMPARISON OPERATORS

99. What is a non-equi join?

1. A join using conditions other than equality such as < or >
2. A join without tables
3. A join that removes NULLs
4. A join that only works with strings

Answer 1 · A join using conditions other than equality such as < or >

PERFORMANCE TUNING

100. Why is selecting only needed columns often better than using SELECT *?

1. It can reduce unnecessary data transfer and improve readability
2. It automatically creates indexes
3. It guarantees perfect performance
4. It removes duplicates

Answer 1 · It can reduce unnecessary data transfer and improve readability

PERFORMANCE TUNING

101. Why can filtering earlier in a query help performance?

1. It reduces the amount of data processed in later steps
2. It always changes NULLs to zero
3. It removes the need for joins
4. It sorts data automatically

Answer 1 · It reduces the amount of data processed in later steps

PERFORMANCE TUNING

102. Why can functions applied directly to filtered columns sometimes hurt performance?

1. They may make it harder for the database to use indexes efficiently
2. They always return NULL
3. They force a FULL OUTER JOIN
4. They remove GROUP BY support

Answer 1 · They may make it harder for the database to use indexes efficiently

PERFORMANCE TUNING

103. What is a common tradeoff when using many nested subqueries instead of clearer CTEs or staged logic?

1. Reduced readability and harder debugging
2. Guaranteed faster execution
3. Automatic duplicate removal
4. Permanent storage of results

Answer 1 · Reduced readability and harder debugging

SQL LOGIC

104. Why is understanding operator precedence important in WHERE clauses?

1. Because AND and OR may be evaluated in an order different from what you intended
2. Because SQL ignores parentheses
3. Because precedence only matters in ORDER BY
4. Because precedence changes column names

Answer 1 · Because AND and OR may be evaluated in an order different from what you intended

SUBQUERIES VS JOINS

105. In many analytical cases, why might a JOIN be preferred over a subquery?

1. It can be easier to read and optimize depending on the problem
2. It always returns fewer rows
3. It avoids all duplicates
4. It never needs keys

Answer 1 · It can be easier to read and optimize depending on the problem

WINDOW FUNCTIONS

106. Which use case best fits LAG() or LEAD()?

1. Comparing each row to a previous or next row
2. Counting all rows in a table
3. Removing duplicates permanently
4. Creating a temp table

Answer 1 · Comparing each row to a previous or next row

SQL ANALYSIS

107. Which query pattern is best for finding the top-paid employee in each department while still showing each employee row?

1. A window function with PARTITION BY department and ORDER BY salary DESC
2. A simple COUNT(*)
3. A DISTINCT on salary only
4. A UNION of all departments

Answer 1 · A window function with PARTITION BY department and ORDER BY salary DESC

SQL ANALYSIS

108. Why is HAVING COUNT(*) > 1 commonly used in duplicate checks?

1. It filters grouped values that appear more than once
2. It sorts duplicates alphabetically
3. It removes duplicates automatically
4. It converts NULLs to zeros

Answer 1 · It filters grouped values that appear more than once

SQL BASICS

109. Which SQL clause is used to choose specific columns from a table?

1. SELECT
2. FROM
3. WHERE
4. ORDER BY

Answer 1 · SELECT

SQL BASICS

110. Which clause tells SQL which table to read data from?

1. WHERE
2. SELECT
3. FROM
4. GROUP BY

Answer 3 · FROM

SQL BASICS

111. What does the LIMIT clause do in a SQL query?

1. Sorts rows
2. Filters columns
3. Restricts number of returned rows
4. Combines tables

Answer 3 · Restricts number of returned rows

ORDER BY

112. If you write ORDER BY salary DESC, how will the rows be sorted?

1. Lowest to highest salary
2. Alphabetically by salary
3. Highest to lowest salary
4. Random order

Answer 3 · Highest to lowest salary

ARITHMETIC OPERATORS

113. Which expression correctly increases a price column by 10?

1. price + 10
2. price ADD 10
3. INCREASE(price, 10)
4. price PLUS 10

Answer 1 · price + 10 (answer reconstructed - lost in source file)

LIKE AND IN

114. Which operator is best for matching text patterns?

1. BETWEEN
2. LIKE
3. IN
4. COUNT

Answer 2 · LIKE

LIKE AND IN

115. Which operator is best when checking whether a value matches one of several listed values?

1. LIKE
2. IN
3. ORDER BY
4. HAVING

Answer 2 · IN

BETWEEN

116. What does BETWEEN usually do in SQL?

1. Matches patterns
2. Checks whether a value falls within a range
3. Counts rows
4. Joins two tables

Answer 2 · Checks whether a value falls within a range

NOT AND OR

117. Which operator reverses a condition in SQL?

1. NOT
2. OR
3. AND
4. IN

Answer 1 · NOT

JOIN BASICS

118. What is the main purpose of a JOIN in SQL?

1. To sort rows
2. To combine related data from multiple tables
3. To delete duplicates
4. To rename columns

Answer 2 · To combine related data from multiple tables

INNER JOIN

119. An INNER JOIN returns:

1. All rows from both tables
2. Only matching rows between tables
3. Only rows from the left table
4. Only rows with NULL values

Answer 2 · Only matching rows between tables

AGGREGATION

120. Which function counts rows or non-null values in SQL?

1. SUM
2. COUNT
3. AVG
4. MIN

Answer 2 · COUNT

AGGREGATION

121. Which function adds numeric values together?

1. COUNT
2. SUM
3. AVG
4. MAX

Answer 2 · SUM

AGGREGATION

122. Which function returns the average of numeric values?

1. AVG
2. COUNT
3. MIN
4. DISTINCT

Answer 1 · AVG

AGGREGATION

123. Which clause is used to group rows before applying aggregate functions?

1. ORDER BY
2. GROUP BY
3. WHERE
4. LIMIT

Answer 2 · GROUP BY

NULLS

124. Which statement about NULL is correct?

1. NULL equals 0
2. NULL means an empty string
3. NULL represents missing or unknown data
4. NULL is always false

Answer 3 · NULL represents missing or unknown data

CASE

125. What is the purpose of a CASE statement in SQL?

1. To join tables
2. To create conditional logic in query output
3. To sort rows
4. To count duplicates

Answer 2 · To create conditional logic in query output

CTES

126. What is a key advantage of a Common Table Expression CTE?

1. It permanently stores data
2. It improves readability by breaking logic into steps
3. It replaces all joins
4. It only works with ORDER BY

Answer 2 · It improves readability by breaking logic into steps

TEMP TABLES

127. How is a temporary table generally different from a CTE?

1. A temp table can be reused across multiple queries in a session
2. A temp table cannot store rows
3. A CTE is permanent
4. A CTE always performs faster

Answer 1 · A temp table can be reused across multiple queries in a session

WINDOW FUNCTIONS

128. What is a major benefit of window functions compared with GROUP BY?

1. They delete duplicates
2. They keep row-level detail while performing calculations
3. They only work on text columns
4. They replace WHERE clauses

Answer 2 · They keep row-level detail while performing calculations

WINDOW FUNCTIONS

129. Which clause is commonly used inside a window function to divide rows into groups?

1. HAVING
2. PARTITION BY
3. LIMIT
4. DISTINCT

Answer 2 · PARTITION BY

WINDOW FUNCTIONS

130. Which clause inside a window function helps define row order for running totals or ranking?

1. WHERE
2. ORDER BY
3. FROM
4. IN

Answer 2 · ORDER BY

SQL BASICS

131. Which query correctly selects all columns from a table named customers?

1. SELECT ALL FROM customers
2. SELECT * FROM customers
3. GET * FROM customers
4. FROM customers SELECT *

*Answer 2 · SELECT * FROM customers*

WHERE

132. Which condition correctly filters rows where age is greater than 30?

1. age > 30
2. age => 30
3. age more than 30
4. age <> 30

Answer 1 · age > 30

ORDER BY

133. What is the default sort direction for ORDER BY if ASC or DESC is not specified?

1. Descending
2. Ascending
3. Random
4. Grouped

Answer 2 · Ascending

LIKE

134. Which pattern matches values that start with the letter A?

1. '%A'
2. 'A%'
3. '%A%'
4. '_A'

Answer 2 · 'A%'

LIKE

135. Which pattern matches values that contain the word data anywhere in the text?

1. data%
2. %data
3. %data%
4. _data_

Answer 3 · %data%

IN

136. Which query condition is equivalent to country = 'India' OR country = 'USA'?

1. country BETWEEN 'India' AND 'USA'
2. country IN ('India', 'USA')
3. country LIKE 'India, USA'

Answer 2 · country IN ('India', 'USA') (answer reconstructed - lost in source file)

BETWEEN

137. Which statement about BETWEEN is true?

1. It excludes both endpoints
2. It includes both endpoints
3. It only works for text
4. It is the same as LIKE

Answer 2 · It includes both endpoints

AND OR

138. What is the effect of OR in a WHERE clause?

1. Both conditions must be true
2. At least one condition must be true
3. It reverses the condition
4. It groups rows

Answer 2 · At least one condition must be true

NOT

139. Which condition finds rows where status is not 'closed'?

1. status != 'closed'
2. status NOT 'closed'
3. NOT status closed
4. status <> closed

Answer 1 · status != 'closed'

JOIN BASICS

140. Which column is typically used in a JOIN condition?

1. Any random column
2. A related key shared between tables
3. Only text columns
4. Only aggregated columns

Answer 2 · A related key shared between tables

INNER JOIN

141. If a customer has no matching order, what happens to that customer row in an INNER JOIN between customers and orders?

1. It always appears with NULLs
2. It is excluded
3. It is duplicated
4. It becomes a temp table

Answer 2 · It is excluded

COUNT

142. What is the difference between COUNT(*) and COUNT(column_name)?

1. No difference ever
2. COUNT(column_name) ignores NULLs while COUNT(*) counts rows
3. COUNT(*) ignores duplicates only
4. COUNT(column_name) counts tables

Answer 2 · COUNT(column_name) ignores NULLs while COUNT(*) counts rows

SUM AND AVG

143. Which statement is true about SUM and AVG?

1. They only work on text
2. SUM adds values and AVG computes the mean
3. SUM sorts values and AVG filters rows
4. AVG counts rows

Answer 2 · SUM adds values and AVG computes the mean

MIN MAX

144. Which pair of functions returns the smallest and largest values in a column?

1. LOW and HIGH
2. FIRST and LAST
3. MIN and MAX
4. START and END

Answer 3 · MIN and MAX

GROUP BY

145. Why would you use GROUP BY department?

1. To sort departments alphabetically
2. To calculate aggregates for each department
3. To remove NULLs
4. To rename the department column

Answer 2 · To calculate aggregates for each department

HAVING

146. Why can't HAVING always be replaced by WHERE?

1. WHERE works after aggregation and HAVING works before aggregation
2. HAVING filters groups after aggregation
3. WHERE only sorts rows
4. HAVING only works with joins

Answer 2 · HAVING filters groups after aggregation

DISTINCT

147. Which query is best for listing unique job titles from an employees table?

1. SELECT job_title FROM employees
2. SELECT UNIQUE job_title FROM employees
3. SELECT DISTINCT job_title FROM employees
4. SELECT ONLY job_title FROM employees

Answer 3 · SELECT DISTINCT job_title FROM employees

CASE

148. What will a CASE expression usually return?

1. A new calculated value based on conditions
2. A new table
3. A permanent column in the database
4. A join key

Answer 1 · A new calculated value based on conditions

DATE FUNCTIONS

149. Why are DATE functions useful in SQL?

1. They only format text
2. They help extract or compare parts of dates for analysis
3. They replace GROUP BY
4. They are only for deleting old rows

Answer 2 · They help extract or compare parts of dates for analysis

SUBQUERIES

150. When is a scalar subquery most appropriate?

1. When you need a single value inside another query
2. When joining five tables
3. When creating an index
4. When sorting multiple columns

Answer 1 · When you need a single value inside another query

SUBQUERIES IN WHERE

151. Why use a subquery in a WHERE clause?

1. To rename columns
2. To filter rows based on the result of another query
3. To sort grouped rows
4. To create a permanent view

Answer 2 · To filter rows based on the result of another query

CTES

152. Which statement about CTEs is correct?

1. They are always stored permanently
2. They exist only for the duration of a single query
3. They cannot include joins
4. They replace SELECT

Answer 2 · They exist only for the duration of a single query

TEMP TABLES

153. When are temporary tables especially useful?

1. When the same intermediate result is needed repeatedly
2. When selecting one column only
3. When avoiding all joins
4. When writing a simple LIMIT query

Answer 1 · When the same intermediate result is needed repeatedly

WINDOW FUNCTIONS

154. Which task is especially well suited to a window function?

1. Deleting duplicate rows
2. Calculating a running total by date
3. Renaming a table
4. Creating a database

Answer 2 · Calculating a running total by date

WINDOW FUNCTIONS

155. Why might an analyst prefer a window function over a subquery for some comparisons?

1. Because window functions can show row detail and comparison metrics together
2. Because subqueries cannot filter
3. Because window functions only work on dates
4. Because subqueries are invalid SQL

Answer 1 · Because window functions can show row detail and comparison metrics together

DATA CLEANING

156. Which issue is commonly addressed during SQL data cleaning?

1. Adding more tables
2. Handling duplicates and NULLs
3. Changing SQL into Python
4. Removing all numeric columns

Answer 2 · Handling duplicates and NULLs

PERFORMANCE TUNING

157. Which practice generally helps keep SQL easier to maintain?

1. Writing everything in one giant query with no structure
2. Using clear logic with tools like CTEs when appropriate
3. Avoiding aliases always
4. Using DISTINCT in every query

Answer 2 · Using clear logic with tools like CTEs when appropriate